Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[12EE82]

EIGHTH SEMESTER B.E EXAMINATION, JUNE - 2016

RENEWABLE ENERGY RESOURCES

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	What are the advantages of renewable energy?	04
	b.	Explain the main obstacles to the implementation of renewable energy systems.	05
	c.	In detail mention the prospects of renewable energy sources.	05
2.	a.	Explain in detail solar radiation at the earth's surface with different points of irradiations.	08
	b.	What is Pyranometer? With neat diagram describe the function of Eppley pryanometer.	06
3.	a.	What principle been adopred in the conversion of solar radiation to heat? Explain in detail.	08
	b.	With a neat diagram explain the working principle of solar furnaces and state advantages and limitations.	08
4.	a.	With a neat diagram describe the operating principle of non-convective solar pond stating its limitations on the characteristics.	08
	b.	Enumerate solar cell principles and explain the basic working knowledge of typical solar cells.	06
5.	a.	With a block diagram explain the working of WECS.	08
	b.	What are the points considered for the wind site selection?	06
6.	a.	Describe the Dry process methodology in the biomass conversion technology.	08
	b.	What are the different factors affecting bio digestion or generation of gas.	06

7.	**a.**	With neat schematic diagram explain the process of open cycle OTEC (ocean thermal energy conversion) in detail.	**06**
	b.	Estimate the energy and power in simple single basics tidal system and double cycle system.	**08**
8.	**a.**	With a block diagram explain the working of open cycle MHD system.	**08**
	b.	Explain in detail V-I, characteristics of fuel cell with energy equation.	**06**